

Capstone Project Proposal

CSE 598 · Agentic AI · Fall 2026

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Project	Claim Denial Triage Agent for Outpatient Revenue Cycle
Repository	https://github.com/Mokshitha1303/claim-denial-triage
Configuration	examples/readme.md

1. Problem Definition

Task	Given one denied medical claim, decide why it was denied, what to do about it, and by when.
User	An accounts-receivable specialist at an outpatient practice, working a denial worklist.
Input	A claim record in JSON: payer, dates, service lines, the denial codes returned, and encounter facts such as whether an authorization is on file.
Output	Root cause, disposition, required actions, filing deadline and confidence — plus a draft appeal letter when the denial is appealable.
Success	Root cause and disposition match a trained biller's call, the actions name the actual defect, and the deadline is right for that payer.
Failure	Wrong root cause. An appeal on a contractual adjustment. A missed deadline.

2. Motivation and Project Scope

Denials are a steady revenue leak for outpatient practices, and a large share are never reworked. That is usually a triage problem, not a merit problem — by the time anyone works out that a claim was recoverable, the filing window has closed.

It suits an agent because one decision needs several lookups — the code set, the payer's contract, its medical policy, the encounter record — and which comes next depends on what the denial turns out to be. Planning, tool use and retrieval, ending in a draft behind a human approval gate.

In scope this semester

- Professional (CMS-1500) outpatient claims
- Ten root-cause categories
- Disposition, action plan and deadline
- Appeal drafting from retrieved payer policy
- An evaluation harness, and a human approval gate

Deliberately out of scope

- Institutional (UB-04) claims
- Live clearinghouse or EHR integration
- Any automated submission to a payer
- Real patient data — everything is synthetic

3. Runnable Baseline

Stack. Python 3.9+, standard library only. No model call, no network, no API key, no install step.

1. Load and validate the claim, so malformed input never reaches the rules.
2. Map each CARC to a root-cause category, using RARC codes to pin down the field the payer objected to.
3. Load the payer's contract profile — filing windows, appeal name, submission channel.
4. Apply that category's rule, branching on encounter facts. CO-197 with an auth on file is a resubmission; with none, an escalation.
5. Compute the deadline, and force manual review if it has already passed.
6. Draft an appeal letter — for appealable findings only.
7. Render the report and the structured record.

Why start here. Triage has a large deterministic core — code mapping, group-code semantics, deadline arithmetic — where a model adds cost and variance but not accuracy. Rules fix the decision vocabulary first and make the residue visible: the claims this baseline escalates are the ones needing retrieval and chart reading.

Where it lives. `run_baseline.py` drives `triage/` — validation, lookups, rules, letter, report — with reference data in `data/`, six labeled claims in `examples/`, nine tests in `tests/`.

4. Test Case and Baseline Output

Input	examples/test2_medical_necessity.json — a brain MRI (CPT 70553, \$1,450) billed to BCBS AZ, denied CO-50 with remark M127 . A clinical note is on file.
Expected	Read it as a medical necessity denial, mark it appealable, compute the 180-day BCBS AZ deadline, and draft a letter enclosing the documentation on file.

```
python run_baseline.py --input examples/test2_medical_necessity.json --as-of 2026-09-06
```

```
claim-denial-triage - python

$ python run_baseline.py --input examples/test2_medical_necessity.json --as-of 2026-09-06

PRIMARY DISPOSITION : APPEAL_WITH_DOCUMENTATION
PRIMARY ROOT CAUSE   : MEDICAL_NECESSITY

Disposition : APPEAL_WITH_DOCUMENTATION (confidence: high)
Deadline    : 2027-01-07 (123 days remaining)

Required actions:
1. Locate the Blue Cross Blue Shield of Arizona medical policy for CPT 70553
   and map the chart findings to its coverage criteria.
2. Attach clinical_note, progress_notes to the appeal.
3. File a Provider Claim Dispute via Availity provider portal.

[ full report and draft appeal letter in the appendix ]
```

What worked. Routing is correct, and the deadline and appeal name come from the payer's contract profile rather than being hardcoded. All six example claims match their labels, in under a second.

What did not. The first action is the tell — the baseline can say *find the medical policy*, but it cannot find, read, or check it against the chart. On a bare CO-16 it refuses to guess and escalates: right behavior, no dollars recovered.

5. Reproducibility and Run Instructions

Dependencies: Python 3.9 or newer, nothing else. **API keys or environment variables:** none.

```
git clone https://github.com/Mokshitha1303/claim-denial-triage.git
cd claim-denial-triage
python run_baseline.py --input examples/test2_medical_necessity.json
```

Input	examples/*.json, with the schema documented in examples/readme.md
Output	outputs/<claim_id>.txt and .json, and the terminal
All six	python run_baseline.py --input examples/
Tests	python -m unittest discover -s tests -v → Ran 9 tests ... OK
Caveat	Deadlines count from today; pass --as-of 2026-09-06 to reproduce the run above exactly.

6. Initial Evaluation Plan

The harness grows out of the existing label file, extended to about fifty claims. The agent earns its keep if it decides a real share of the claims this baseline escalates, without regressing on the ones it already gets right.

- **Root cause and disposition accuracy** — exact match against biller labels. The headline metric.
- **Action specificity** — do the actions name the real defect, or just restate the denial?
- **Escalation calibration** — over-escalation against under-escalation; the second is the worse error.
- **Letter groundedness** — LLM-as-a-judge plus spot review against the retrieved policy and the chart.
- **Cost, latency, tool failure rate**, and safety counts that must stay at zero: no payer submission.

7. Limitations and Next Steps

Weaknesses	Sixteen CARC and nine RARC codes are mapped, and filing windows are illustrative defaults. Encounter facts are booleans, so it never reads clinical text.
Likely failures	Bare CO-16 denials. Codes outside the local set. Payer policy nuance. Multi-line claims where denials interact.
Next phase	Turn the rules into tools an agent plans over — code lookup, a retriever over payer medical policy, an encounter fetch, an eligibility check — so it reads the note against retrieved criteria and writes a grounded appeal.
Risks and help	No public labeled denial dataset exists, so label quality is the main threat. I would benefit from a reviewer with revenue-cycle experience and a set of public payer policies.

Appendix · Full baseline run

Complete terminal output for the command in Section 4, including the generated draft appeal letter.

```
claim-denial-triage - python

$ python run_baseline.py --input examples/test2_medical_necessity.json --as-of 2026-09-06

=====
CLAIM DENIAL TRIAGE REPORT
=====
Claim ID       : CLM-2026-005097
Patient Account : PA-90455
Payer          : Blue Cross Blue Shield of Arizona (BCBS_AZ)
Date of Service : 2026-06-02
Remittance Date : 2026-07-11
Evaluated On    : 2026-09-06

Billed $1,450.00  Paid $0.00  Denied $1,450.00  Recoverable $1,450.00

PRIMARY DISPOSITION : APPEAL_WITH_DOCUMENTATION
PRIMARY ROOT CAUSE  : MEDICAL_NECESSITY

=====
FINDING 1 - line 1 - CPT 70553 - $1,450.00
=====
Denial code : CO-50 RARC M127
Group meaning : Contractual Obligation - provider writes off, cannot bill the patient
Payer says : Non-covered service because the payer does not deem it a medical necessity.
Root cause : MEDICAL_NECESSITY
Disposition : APPEAL_WITH_DOCUMENTATION (confidence: high)
Deadline : 2027-01-07 (123 days remaining)
Why : Medical necessity denials are overturned on clinical evidence, not on claim corrections.

Required actions:
1. Locate the Blue Cross Blue Shield of Arizona medical policy for CPT 70553 and map the chart findings to its coverage criteria.
2. Attach clinical_note, progress_notes to the appeal.
3. File a Provider Claim Dispute via Availity provider portal.

=====
DRAFT APPEAL LETTER
=====

[PRACTICE LETTERHEAD]

September 06, 2026

Blue Cross Blue Shield of Arizona
Attn: Provider Claim Dispute Unit

RE: Request for Provider Claim Dispute

Claim ID: CLM-2026-005097
Patient Account: PA-90455
Date of Service: 2026-06-02
Remittance Date: 2026-07-11
Billed Amount: $1,450.00
Amount in Dispute: $1,450.00

To the review team,

The remittance dated 2026-07-11 returned CO-50 on service line 1 (CPT 70553): "Non-covered service because the payer does not deem it a medical necessity."

The service was ordered and performed to address a documented clinical presentation. The enclosed record establishes the indication, the findings, and the treatment decision, and we ask that it be reviewed against the plan's coverage criteria for this procedure.

Enclosures:
- clinical note
- progress notes

Please reprocess the claim and issue a corrected remittance. The billing office is available for any additional information required.

Sincerely,

[NAME], [TITLE]
[PRACTICE NAME] - Revenue Cycle

-- DRAFT: requires biller review and signature before submission --

=====
End of report. Human review required before any payer submission.
=====
Processed 1 claim(s). Output written to outputs/
```

All claim data in this project is synthetic. No protected health information was used. CARC and RARC are public X12 code sets.